SECURE FILE SHARING SYSTEM USING VISUAL CRYPTOGRAPHY AND BLOCKCHAIN

A Project Report submitted in partial fulfillment of the requirements

for the award of the degree of

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Applications

Submitted by

Chintu Madan Sorte
MC240105CA

Under the Guidance of

Prof. Rohan B. Kokate

Education to Eternity

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Year 2025-26

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We hereby declare that the work presented in this project report entitled, **"Secure File Sharing Using Visual Cryptography and Blockchain"** in the subject **Computer Applications** in the faculty of Science and Technology is the original contribution carried out by us under the guidance of **Prof. Rohan B. Kokate Department of Computer Applications, J D College of Engineering and Management, Nagpur.** This work has not been submitted to any other University or Institution for the award of any degree or diploma or certificate course.

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Date:

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This is to certify that the project report entitled, **"Secure File Sharing Using Visual Cryptography and Blockchain"** in the subject **Computer Applications** in the faculty of Science and Technology submitted by **Chintu Madan Sorte** to **Dr. Babasaheb Ambedkar Technological University, Lonere** for the award of the degree of **Master of Computer Application** is a Bonafide record of work carried out by them under my supervision. The contents of this Project Report, in full or in parts, have not been submitted or published to any other Institute or University for the awardof any degree or diploma.

Prof. Rohan B. Kokate
MCA

Forwarded to:

Prof. Sudhanshu Tripathi
Project Coordinator

Prof. Rohan B. Kokate
Head Of the Department
CA

Dr. S. V. Sonekar
Principal

CERTIFICATE OF APPROVAL

This is to certify that the Project Report on **Secure File Sharing Using Visual Cryptography and Blockchain** is approvedwork done by **Chintu M. Sorte** in partial fulfillment of the requirements for the award of the degree of **Master of Computer Applications** at J D College of Engineering & Management, Nagpur affiliated to **Dr. Babasaheb Ambedkar Technological University, Lonere during** the academic year 2025-26.

Prof. Rohan B. Kokate
Guide

Prof. Rohan B. Kokate
Head of the Department

Project Examination held on _____

Internal Examiner/ Guide

External Examiner

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ABBREVIATIONS

VC	Visual Cryptography
SSS	Shamir's Secret Sharing
IPFS	Interplanetary File System
API	Application Programming Interface
CID	Content Identifier
UI	User Interface
UX	User Experience
DB	Database
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
JWT	JSON Web Token
SHA	Secure Hash Algorithm
P2P	Peer-to-Peer
Dapp	Decentralized Application
EVM	Ethereum Virtual Machine
RPC	Remote Procedure Call

ABSTRACT

In the current digital landscape, secure file sharing has become a critical requirement due to the increasing risk of data breaches, unauthorized access, and centralized system failures. Traditional file sharing methods often rely on single-point storage and encryption techniques that are vulnerable to attacks and data manipulation. To address these challenges, this project proposes a secure and decentralized file sharing system that integrates Visual Cryptography, Blockchain technology, and Interplanetary File System (IPFS).

The proposed system enhances data security by dividing a file into multiple encrypted shares using Visual Cryptography techniques. These shares are individually meaningless and reveal the original file only when combined, ensuring confidentiality even if some shares are compromised. A dedicated Python-based Visual Cryptography service is used to perform share generation and reconstruction efficiently.

To ensure data integrity and secure access control, Blockchain technology is incorporated to store cryptographic hashes of file shares and enforce access permissions through smart contracts. This prevents unauthorized modifications and guarantees transparency in file access. Additionally, IPFS is used for decentralized storage of file shares, eliminating reliance on centralized servers and improving system robustness.

The system architecture consists of a frontend interface for user interaction, a backend for API management and orchestration, a Visual Cryptography service for processing, Blockchain for verification, and IPFS for storage. Together, these components provide a multi-layered security model that ensures confidentiality, integrity, and availability of shared data.

The results demonstrate that the proposed system effectively secures file sharing by combining cryptographic techniques with decentralized technologies. It offers improved resistance against attacks, enhanced data privacy, and a scalable approach suitable for modern secure communication systems.

CHAPTER 1: INTRODUCTION

1.1 BRIEF OUTLINE OF PROJECT

In today's digital era, the exchange of information through online platforms has become an essential part of personal, academic, and professional activities. With the rapid growth of cloud storage systems and communication technologies, file sharing has become faster and more convenient. However, this convenience has also introduced significant security concerns, including unauthorized access, data breaches, and loss of confidentiality.

Secure file sharing is a critical requirement in modern systems, especially when sensitive information such as personal data, financial records, or confidential documents is involved. Traditional file-sharing methods primarily rely on encryption and centralized storage systems. While encryption provides a certain level of security, centralised systems remain vulnerable to attacks, single points of failure, and data tampering.

To address these limitations, there is a growing need for advanced security mechanisms that ensure the confidentiality, integrity, and availability of data. One such approach is Visual Cryptography (VC), a technique that divides an image or file into multiple shares. These shares appear meaningless individually and reveal the original content only when combined. This method eliminates the need for complex decryption algorithms and enhances data security.

Furthermore, the use of decentralised storage systems, such as the Interplanetary File System (IPFS), eliminates reliance on centralised servers. IPFS distributes data across a network, improving reliability, fault tolerance, and resistance to censorship or system failures.

This project proposes a secure file-sharing system that integrates Visual Cryptography, Blockchain, and IPFS, along with Firebase for authentication and backend support. The system divides files into secure shares using a Python-based VC service, stores them in decentralised storage, and uses blockchain for verification and access control. This multi-layered architecture enhances overall system security and ensures safe and efficient file sharing.

1.2 OVERVIEW OF PROJECT REPORT

This report is organized into multiple chapters, each focusing on a specific aspect of the project. The structure of the report is designed to provide a clear understanding of the problem, methodology, implementation, and results.

Chapter 1 introduces the concept of secure file sharing, highlights existing challenges, and presents the proposed solution along with its key components.

Chapter 2 provides a detailed literature survey, including a review of existing research in visual cryptography, blockchain, and decentralized storage systems. It also identifies research gaps, defines the problem statement, and outlines the objectives of the project.

Chapter 3 explains the research methodology and system design. It describes the architecture of the system, the technologies used, and the workflow of different modules including the frontend, backend, VC service, blockchain, and storage components.

Chapter 4 focuses on the implementation details of the system. It explains how each module is developed and how they interact with each other to achieve secure file sharing.

Chapter 5 presents the results obtained from the system along with analysis and discussion. It evaluates the performance, security, and effectiveness of the proposed solution.

Chapter 6 concludes the report by summarizing the work done, highlighting key findings, and suggesting possible improvements and future enhancements.

Overall, this report demonstrates the development of a secure, decentralized, and efficient file sharing system by combining modern cryptographic and distributed technologies.

CHAPTER 2: LITERATURE SURVEY

2.1 LITERATURE REVIEW

The field of secure file sharing has evolved significantly with advancements in cryptography, distributed systems, and decentralized technologies. This section reviews the major concepts and technologies relevant to the proposed system.

Study of Visual Cryptography

Visual Cryptography (VC) is a technique that allows secure sharing of visual information by dividing it into multiple shares. These shares individually do not reveal any meaningful information, but when combined, they reconstruct the original data. The method eliminates the need for complex decryption algorithms and relies on simple superimposition. Over time, VC has been extended beyond images to secure file and data sharing using schemes such as threshold-based secret sharing.

Review of Secure File Sharing Techniques

Traditional secure file sharing systems primarily use encryption algorithms such as symmetric and asymmetric cryptography. While these methods provide confidentiality, they depend heavily on secure key management. Many systems rely on centralized servers, which introduce risks such as data breaches, unauthorized access, and system downtime. Cloud-based systems improve accessibility but still face trust and control issues.

Blockchain in Data Security

Blockchain technology provides a decentralized and tamper-resistant ledger system. It ensures data integrity by storing cryptographic hashes in blocks that are linked together. Once data is recorded, it cannot be modified without consensus from the network. In secure file sharing, blockchain is used for access control, verification, and maintaining an immutable record of transactions. Smart contracts further automate permission management and enhance system transparency.

IPFS in Distributed Storage Systems

Interplanetary File System is a peer-to-peer distributed storage system that allows data to be stored across multiple nodes instead of a single centralized server. Files are identified using cryptographic hashes, ensuring data integrity and efficient retrieval. IPFS improves fault tolerance, reduces dependency on central authorities, and enhances availability of stored data.

Existing Hybrid Security Models

Recent research has explored combining multiple technologies such as cryptography, blockchain, and distributed storage to enhance security. Hybrid models aim to address the limitations of individual systems by integrating their strengths. However, many existing solutions lack proper coordination between components or fail to provide a complete end-to-end secure architecture.

FIGURE 2.1: COMPARATIVE ANALYSIS – TRADITIONAL vs PROPOSED SYSTEM

ASPECT	TRADITIONAL FILE SHARING SYSTEMS	PROPOSED SYSTEM (VC + BLOCKCHAIN + IPFS + FIREBASE)
Storage	Centralized servers (single point of failure)	Decentralized storage using IPFS (no single point of failure)
Security Mechanism	Relies on encryption alone; vulnerable to key leakage or data breaches	Multi-layer security using Visual Cryptography, Blockchain, and decentralized storage
Access Control	Managed by central authority; prone to unauthorized access	Smart contracts on Blockchain ensure transparent and tamper-proof access control
Data Integrity	Data can be altered or deleted by malicious actors	Blockchain stores cryptographic hashes ensuring immutability and integrity
Availability	Downtime possible if central server fails or is attacked	High availability through decentralized nodes in IPFS network
Trust Model	Requires trust in central server/provider	Trustless environment using Blockchain and distributed storage
Scalability	Limited scalability due to server capacity constraints	Highly scalable with decentralized architecture and modular components
Transparency	Limited transparency; actions are not publicly verifiable	All transactions and access logs are recorded on Blockchain (transparent & auditable)

Figure 2.1 Comparative analysis – Traditional vs Proposed System

2.2 RESEARCH GAP

Despite significant advancements, several limitations still exist in current systems:

- There is limited integration between Visual Cryptography and Blockchain technologies for secure file sharing.
- Most existing systems rely on centralized storage, making them vulnerable to attacks and failures.
- Access control mechanisms are often inefficient or depend on trusted third parties.
- Many solutions lack a multi-layered security approach that combines confidentiality, integrity, and availability in a unified system.

These gaps highlight the need for a more robust and integrated solution.

2.3 PROBLEM STATEMENT

With the increasing dependency on digital platforms, ensuring secure file sharing has become a major challenge. Existing systems fail to provide complete security due to centralized storage, weak access control, and vulnerability to attacks.

The problem addressed in this project is the design and development of a secure, decentralized file sharing system that:

- Protects sensitive data from unauthorized access
- Ensures data integrity and confidentiality
- Eliminates single points of failure
- Provides reliable and verifiable access control

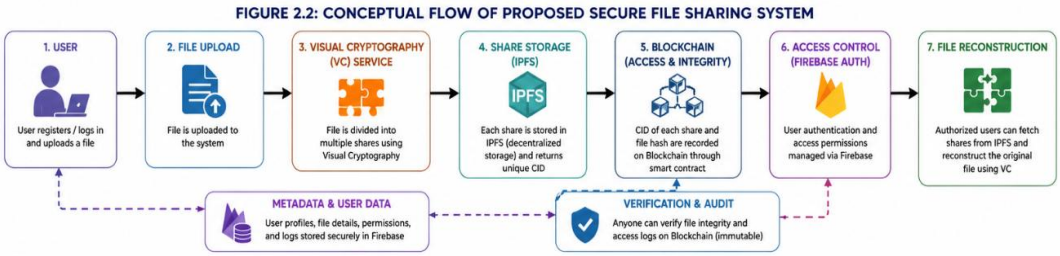

Figure 2.2 Conceptual flow of proposed secure file sharing system

2.4 OBJECTIVES

The main objectives of the proposed system are:

- To develop a secure file sharing system using Visual Cryptography techniques
- To implement Blockchain for secure access control and data verification
- To use IPFS for decentralized and reliable file storage
- To ensure secure generation and reconstruction of file shares
- To design a scalable, efficient, and robust system architecture

CHAPTER 3: RESEARCH METHODOLOGY

3.1 SYSTEM ARCHITECTURE DESIGN

The proposed system is designed as a multi-layered architecture that integrates a client interface, backend services, a Visual Cryptography (VC) module, blockchain for verification, and decentralized storage using IPFS. The objective of this architecture is to ensure secure, reliable, and tamper-proof file sharing by combining cryptographic techniques with distributed technologies.

The system follows a modular approach where each component is responsible for a specific task, ensuring scalability and maintainability. The client interacts with the backend through secure APIs, while the backend orchestrates communication between the VC service, database, blockchain, and IPFS network.

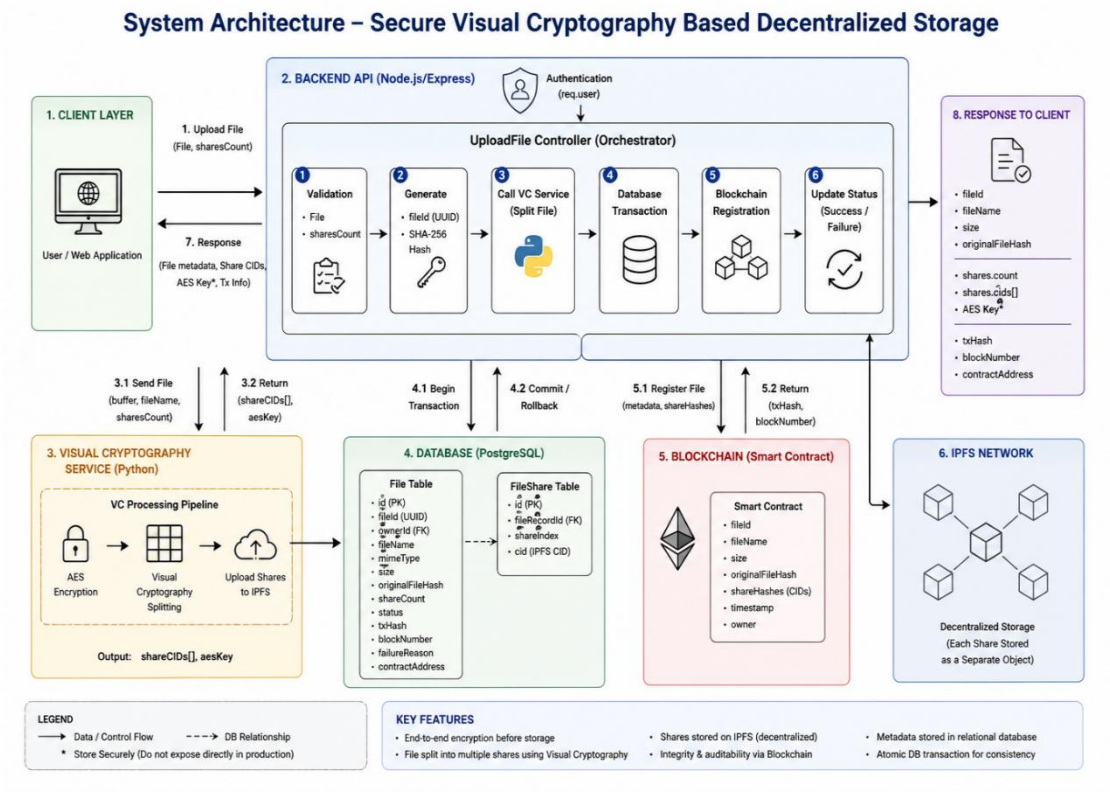

Figure 3.1: System Architecture Diagram

Overall Architecture

The architecture consists of the following major layers:

- **Client Layer:** Handles user interaction such as file upload and retrieval.
- **Backend API Layer:** Acts as the central controller managing requests and coordinating system operations.

- **Visual Cryptography Service:** Processes files into secure shares and reconstructs them when required.
- **Blockchain Layer:** Stores file hashes and manages access control using smart contracts.
- **IPFS Network:** Stores file shares in a decentralized manner.
- **Database Layer:** Maintains metadata, user information, and transaction records.

Data Flow Between Components

The workflow of the system is as follows:

1. The user uploads a file through the client application.
2. The request is sent to the backend API, where validation is performed.
3. A unique identifier and file hash are generated for integrity verification.
4. The backend sends the file to the VC service for splitting.
5. The VC service divides the file into multiple secure shares.
6. These shares are stored in the IPFS network, generating unique content identifiers (CIDs).
7. The backend stores metadata and share details in the database.
8. File hash and share references are recorded on the blockchain through smart contracts.
9. The system returns a response to the user with necessary access details.

During file retrieval:

1. The user requests access to a file.
2. Authentication and permission checks are performed.
3. Shares are fetched from IPFS.
4. The VC service reconstructs the original file.
5. The reconstructed file is returned to the user.

Role of Each Module

- **Client Module:** Provides interface for file upload, download, and user interaction.
- **Backend Module:** Handles business logic, API requests, and integration of all services.
- **VC Service:** Ensures confidentiality by splitting and reconstructing files

securely.

- **Blockchain Module:** Maintains integrity and transparency by storing hashes and access records.
- **IPFS Module:** Provides decentralized storage and ensures high availability of file shares.
- **Database Module:** Stores metadata such as file details, user data, and transaction logs.

3.1.1 VC SERVICE (PYTHON MODULE)

The Visual Cryptography service is a critical component responsible for secure file processing. It ensures that the original file is never stored directly but is divided into multiple secure shares.

File Splitting Using Visual Cryptography / Shamir's Secret Sharing

The uploaded file is processed using Visual Cryptography or threshold-based secret sharing techniques. The file is divided into multiple shares such that only a predefined number of shares can reconstruct the original file. Individual shares do not reveal any meaningful information.

Share Generation Process

- The file is first encrypted for additional security.
- It is then divided into multiple shares using the VC algorithm.
- Each share is encoded and prepared for storage.
- Shares are sent to the backend for storage in IPFS.

Share Reconstruction Mechanism

- Required shares are retrieved from IPFS.
- Shares are combined using the reconstruction algorithm.
- The original file is restored without loss of data.
- The reconstructed file is securely returned to the user.

3.2 TECHNOLOGY & TOOLS USED

The system uses a combination of modern technologies to ensure performance,

scalability, and security.

- **Frontend:** Developed using React with TypeScript and Tailwind CSS for responsive and interactive user interfaces.
- **Backend:** Implemented using Node.js to handle API requests and system orchestration.
- **Blockchain:** Smart contracts deployed on Polygon or local blockchain networks to manage access control and store file hashes.
- **Storage:** IPFS is used for decentralized storage of file shares, while Firebase is used for metadata and additional storage support.
- **Authentication:** Firebase Authentication or custom authentication mechanisms are used to ensure secure user access.

CHAPTER 4: IMPLEMENTATION

4.1 MODULE-WISE IMPLEMENTATION

4.1.1 Frontend Module

The frontend of the system is developed using React with TypeScript and Tailwind CSS. It provides a responsive and user-friendly interface that allows users to interact with the system efficiently. The design focuses on simplicity, usability, and secure interaction with backend services.

File Upload UI

The file upload interface allows users to select and upload files securely. The system performs validation checks such as file type and size before sending the file to the backend. Once validated, the file is transmitted using API calls. The interface also provides feedback to the user regarding upload progress and success or failure status.

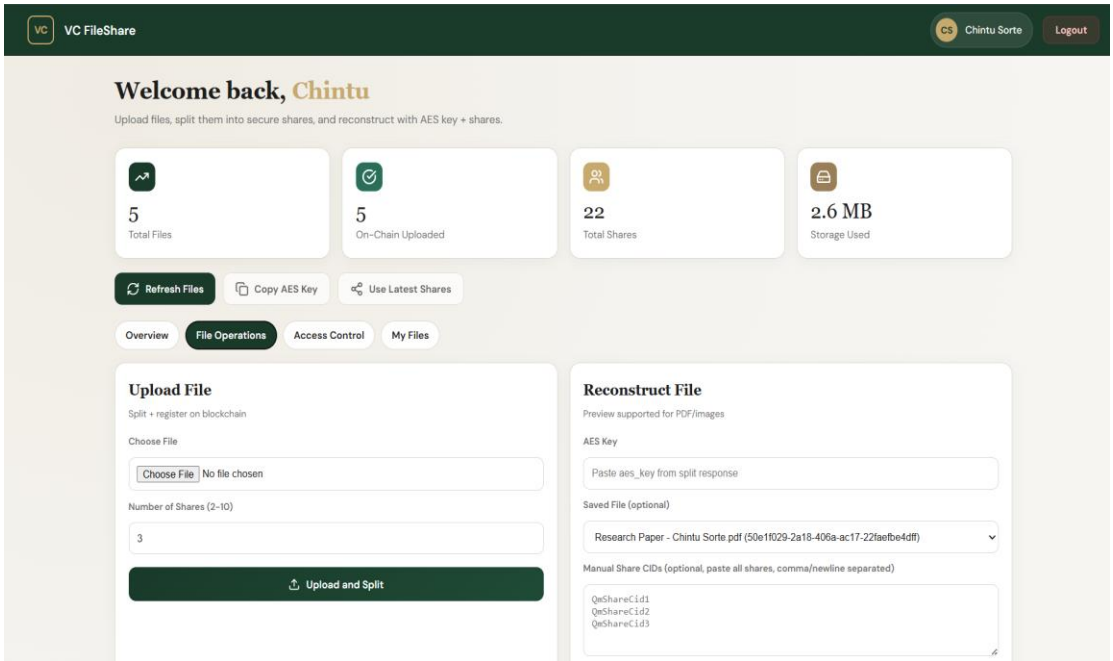

Figure 4.1 File Upload UI

Share Access Interface

The share access interface enables users to retrieve and reconstruct files. It displays available files associated with the user and allows them to initiate reconstruction. The interface communicates with the backend to fetch required shares and ensures that only authorized users can access the data.

User Dashboard

The user dashboard acts as a centralized panel where users can view uploaded files, track their status, and manage access. It improves usability by organizing file-related operations in a structured format. The dashboard also reflects real-time updates from the backend.

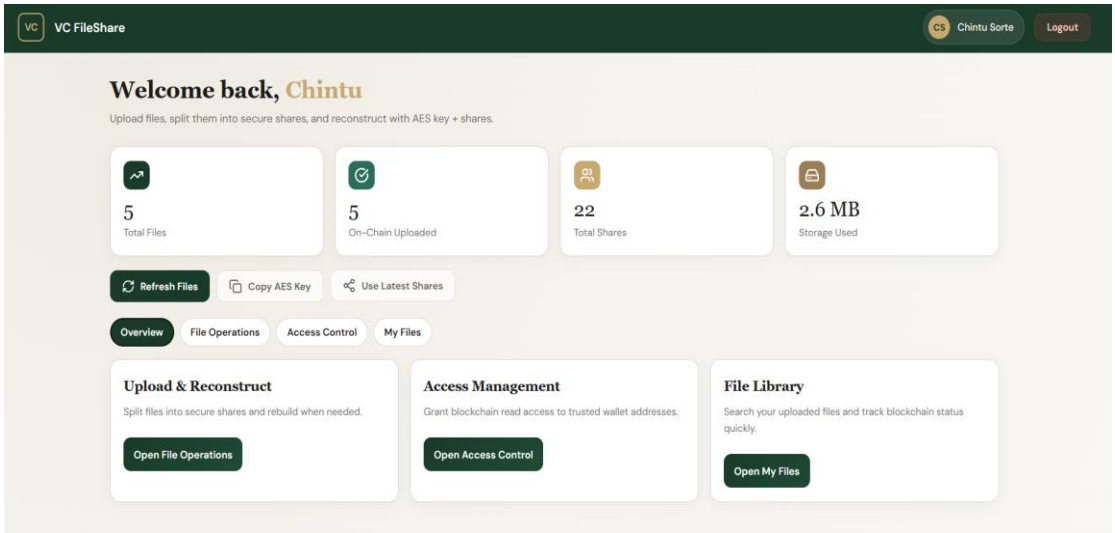

Figure 4.2 User Dashboard UI

4.1.2 Backend Module

The backend is implemented using Node.js and serves as the core controller of the system. It manages API requests, handles business logic, and coordinates communication between different components.

API Design

The system uses RESTful APIs for handling operations such as file upload, retrieval, authentication, and interaction with blockchain and storage systems. Each API endpoint is designed to ensure secure and efficient data transfer.

File Handling & Routing

When a file is uploaded, the backend performs validation and generates a unique identifier along with a cryptographic hash of the file. The file is then routed to the Visual Cryptography service for further processing. The backend also handles error management and ensures proper response delivery.

Integration with VC Service

The backend communicates with the Python-based VC service through internal APIs or service calls. It sends the file for splitting and receives generated shares in return. These shares are then processed for storage and further operations.

4.1.3 VC Service (Python)

The Visual Cryptography service is responsible for securing the file by splitting it into multiple shares and reconstructing it when required.

Share Generation Algorithm

The uploaded file is processed using Visual Cryptography or Shamir's Secret Sharing techniques. The file is divided into multiple shares such that individual shares do not reveal any information. Only a specific combination of shares can reconstruct the original file.

```python
def xor_file_split(data: bytes, n: int):

    shares = []
    length = len(data)

    for _ in range(n - 1):
        shares.append(secrets.token_bytes(length))

    final = bytearray(length)

    for i in range(length):
        v = data[i]
        for s in shares:
            v ^= s[i]
        final[i] = v

    shares.append(bytes(final))

    return shares
```

Figure 4.3 Code Snippet: XOR-based Share Generation

Share Storage Handling

After share generation, the shares are returned to the backend. Each share is then stored in a decentralized storage system such as IPFS. Unique identifiers are generated for each share to enable efficient retrieval.

Reconstruction Logic

```python
def xor_file_reconstruct(shares):

    length = len(shares[0])
    out = bytearray(length)

    for i in range(length):
        v = 0
        for s in shares:
            v ^= s[i]
        out[i] = v

    return bytes(out)
```

Figure 4.4 Code Snippet: XOR-based Reconstruction

When a user requests a file, the required shares are retrieved from storage and passed to the VC service. The service combines these shares using the reconstruction algorithm and restores the original file accurately.

4.1.4 Blockchain Module

The blockchain module ensures data integrity, transparency, and secure access control.

Smart Contract Design

Smart contracts are used to define rules for storing file metadata, including file hashes, ownership details, and access permissions. These contracts automate validation and ensure that only authorized actions are executed.

Hash Storage

The cryptographic hash of each file is stored on the blockchain. This ensures that any modification to the file can be detected, maintaining data integrity.

Access Control Logic

Access control is enforced through blockchain mechanisms. Permissions are verified before allowing file access, ensuring that only authorized users can retrieve or reconstruct files.

4.2 INTEGRATION OF COMPONENTS

The system integrates multiple components to create a secure and efficient file sharing

workflow.

Frontend ↔ Backend Communication

The frontend communicates with the backend through API requests. Operations such as file upload, retrieval, and authentication are handled through these interactions. The communication is designed to be secure and efficient.

Backend ↔ VC Service Interaction

The backend sends files to the VC service for splitting and receives generated shares. It also sends shares for reconstruction when requested. This interaction is crucial for maintaining data confidentiality.

Backend ↔ Blockchain & IPFS

The backend stores file shares in IPFS, which provides decentralized storage. Each share is associated with a unique content identifier. Simultaneously, file hashes and metadata are recorded on the blockchain to ensure integrity and transparency.

End-to-End Workflow Implementation

The complete workflow of the system is as follows:

1. The user uploads a file through the frontend interface.
2. The backend validates the file and generates a unique identifier and hash.
3. The file is sent to the VC service for splitting into shares.
4. The generated shares are stored in IPFS.
5. File metadata and hash are stored on the blockchain.
6. When requested, shares are retrieved from IPFS.
7. The VC service reconstructs the file.
8. The reconstructed file is delivered to the user.

This integrated workflow ensures that the system achieves confidentiality, integrity, and availability while eliminating reliance on centralized storage.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 SYSTEM TESTING

To evaluate the effectiveness of the proposed system, different types of testing were conducted to ensure reliability, security, and performance.

Functional Testing

Functional testing was performed to verify that all system components operate as expected. The testing included file upload, share generation, storage in IPFS, blockchain registration, and file reconstruction.

The system successfully handled file uploads and generated shares without errors. The reconstruction process accurately restored the original file when valid shares were provided. All modules, including frontend, backend, VC service, and blockchain integration, worked cohesively.

Security Testing

Security testing focused on ensuring data confidentiality, integrity, and controlled access. Individual shares generated through the Visual Cryptography process were tested to confirm that they do not reveal any meaningful information independently.

Blockchain-based verification ensured that file hashes remained unchanged, preventing tampering. Access control mechanisms were tested to confirm that unauthorized users were denied access to files.

Performance Testing

Performance testing evaluated the system's efficiency in terms of processing time and responsiveness. The time taken for file upload, share generation, and reconstruction was measured.

Results indicated that the system performs efficiently for moderate file sizes. The use of decentralized storage and modular architecture helps in maintaining scalability, although slight delays were observed due to network-dependent operations such as IPFS retrieval and blockchain transactions.

5.2 RESULTS ANALYSIS

The results obtained from the system demonstrate its effectiveness in achieving secure and reliable file sharing.

File Upload & Share Generation Results

The system successfully processed uploaded files and generated multiple shares using the Visual Cryptography service. Each share appeared as random data and did not reveal any part of the original file. The shares were stored in IPFS and assigned unique identifiers for retrieval.

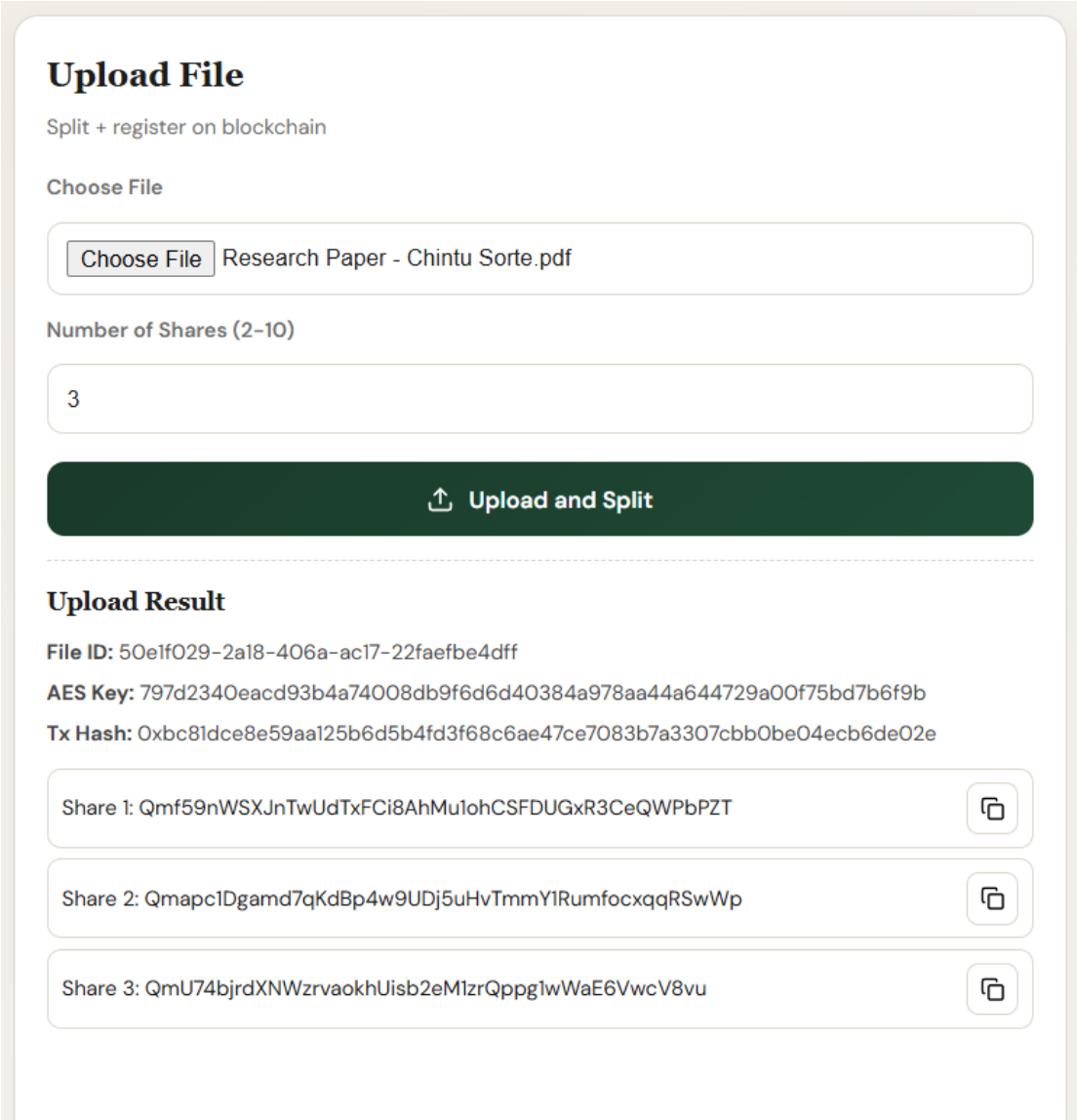

Figure 5.1 File Upload & Share Generation Results

Reconstruction Accuracy

The reconstruction process combined the required shares and restored the original file without any loss or distortion. The output file matched the original file, confirming the correctness of the implemented algorithm.

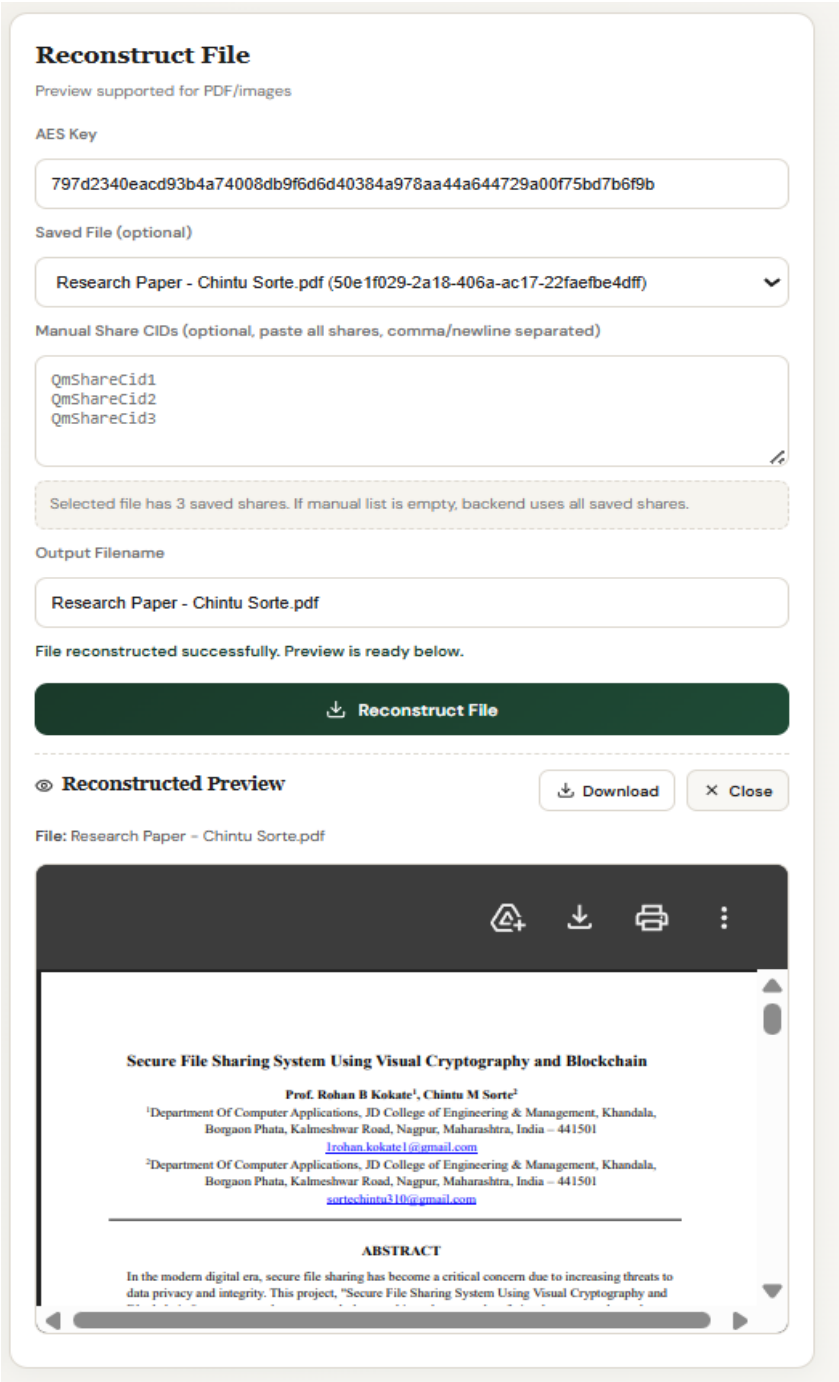

Figure 5.2 File Reconstruction Results

Blockchain Verification Outcomes

Blockchain integration ensured that file hashes were securely stored and remained immutable. During verification, the hash of the reconstructed file matched the hash stored on the blockchain, confirming data integrity. The use of smart contracts enabled secure and transparent access control.

5.3 DISCUSSION

Advantages of Proposed System

- Provides multi-layer security using Visual Cryptography, Blockchain, and IPFS
- Eliminates single point of failure through decentralized storage
- Ensures data integrity using blockchain-based hash verification
- Prevents unauthorized access through controlled permission mechanisms
- Offers scalability due to modular system architecture

Comparison with Existing Systems

Traditional file sharing systems rely on centralized storage and encryption techniques, which are vulnerable to attacks and data breaches. In contrast, the proposed system distributes data across multiple layers, making it more secure and reliable.

Unlike conventional systems, this approach does not store the original file directly. Instead, it divides the file into shares, ensuring that even if one share is compromised, the data remains protected. Additionally, blockchain integration enhances transparency and trust compared to traditional systems.

Limitations of Current Implementation

- Performance depends on network conditions due to IPFS and blockchain interactions
- The current implementation uses a fixed share scheme and does not support dynamic threshold configurations
- Storage and retrieval time may increase with larger file sizes
- System complexity is higher compared to traditional file sharing methods

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5.4 COMPARISON WITH EXISTING SYSTEMS

Metric	Proposed System	Traditional System
Storage	Distributed (IPFS)	Centralized
Security	High (Encryption + Splitting)	Moderate
Integrity	Blockchain-based	Limited
Fault Tolerance	High	Low
Data Privacy	Strong	Moderate

Table 1 Comparison between Proposed System & Traditional System

CHAPTER 6: SUMMARY AND CONCLUSION

6.1 SUMMARY

This project focused on the design and implementation of a secure, decentralized file sharing system by integrating Visual Cryptography (VC), Blockchain technology, and the Interplanetary File System (IPFS). The motivation behind this work was to address the limitations of traditional file sharing systems, which often rely on centralized storage and single-layer security mechanisms, making them vulnerable to data breaches, unauthorized access, and system failures.

The proposed system introduces a multi-layered security approach where the original file is never stored directly. Instead, it is divided into multiple secure shares using Visual Cryptography techniques. These shares are designed in such a way that each individual share appears as random and meaningless data. Only when the required number of shares are combined can the original file be reconstructed. This ensures a high level of confidentiality even if some shares are compromised.

The architecture of the system consists of multiple interconnected components, each responsible for a specific function. The frontend, developed using modern web technologies, provides an intuitive interface for users to upload and access files. The backend, implemented using Node.js, acts as the central controller, managing API requests and coordinating interactions between various modules.

A dedicated Python-based Visual Cryptography service was implemented to handle the splitting and reconstruction of files. This separation of concerns improves modularity and allows the cryptographic logic to operate independently from the main application logic. The generated shares are stored in IPFS, which provides decentralized and distributed storage. Each share is assigned a unique content identifier (CID), ensuring reliable retrieval and data integrity.

To further enhance security, blockchain technology is integrated into the system. Cryptographic hashes of files and share references are stored on the blockchain through smart contracts. This ensures that any tampering with the data can be easily detected. Additionally, blockchain-based access control mechanisms provide a transparent and secure way of managing permissions.

The system was implemented and tested across multiple scenarios, including file

upload, share generation, storage, retrieval, and reconstruction. The results demonstrate that the system performs reliably and maintains data integrity throughout the process. The modular architecture ensures scalability and flexibility, making it suitable for real-world applications.

Overall, this project successfully demonstrates the feasibility of combining cryptographic techniques with decentralized technologies to build a secure and efficient file sharing system.

6.2 CONCLUSION

The implementation of the proposed system highlights the effectiveness of integrating Visual Cryptography, Blockchain, and decentralized storage to address modern data security challenges. Traditional file sharing systems often rely on encryption and centralized servers, which introduce risks such as single points of failure and vulnerability to cyberattacks. In contrast, the proposed system distributes both data and control across multiple layers, significantly improving security and reliability.

Visual Cryptography plays a crucial role in ensuring data confidentiality. By splitting the file into multiple shares, the system ensures that no single entity can access the complete information. Even if an attacker gains access to one or more shares, they cannot reconstruct the original file without the required combination. This approach reduces dependency on complex key management systems and provides an additional layer of protection.

Blockchain technology enhances the system by providing immutability and transparency. The storage of file hashes on the blockchain ensures that any unauthorized modification can be detected immediately. Smart contracts enforce access control policies, ensuring that only authorized users can retrieve and reconstruct files. This eliminates the need for a centralized authority and increases trust in the system.

The use of IPFS for storage further strengthens the architecture by removing reliance on centralized servers. Files are distributed across multiple nodes, improving availability and fault tolerance. Even if some nodes fail, the data can still be retrieved from other nodes in the network.

The system demonstrates strong performance in handling file operations, including upload, share generation, and reconstruction. Although there are slight delays due to network-dependent processes such as IPFS retrieval and blockchain transactions, these

are acceptable trade-offs for the enhanced security and decentralization provided by the system.

In conclusion, the proposed solution successfully achieves its goal of providing a secure, decentralized, and efficient file sharing mechanism. It offers significant improvements over traditional systems and provides a strong foundation for future enhancements. The integration of multiple advanced technologies makes the system robust and adaptable to various application domains.

6.3 SCOPE FOR FUTURE WORK

While the proposed system achieves its intended objectives, there are several areas where further improvements and enhancements can be made to increase its functionality, efficiency, and usability.

Multi-user Threshold Access System

The current implementation primarily supports a fixed share-based reconstruction approach. Future work can extend this to a threshold-based system where a minimum number of shares (k out of n) are required to reconstruct the file. This would provide greater flexibility and allow collaborative access control among multiple users.

AI-based Threat Detection

Security can be further enhanced by integrating artificial intelligence or machine learning techniques. These can be used to monitor user behaviour, detect anomalies, and identify potential threats such as unauthorized access attempts or suspicious activities in real time.

Performance Optimization

Although the system performs efficiently for moderate workloads, optimization techniques can be applied to reduce latency. Improvements in caching, parallel processing, and optimized interaction with IPFS and blockchain networks can enhance overall system performance, especially for large files.

Mobile Application Integration

Developing a mobile application would increase accessibility and usability of the system. A mobile-friendly interface would allow users to securely upload, access, and manage files from anywhere, making the system more practical for real-world use.

Advanced Encryption Layer

While Visual Cryptography provides strong confidentiality, an additional encryption layer can be introduced before share generation. This would further strengthen security and provide protection against advanced attack scenarios.

Improved Share Visibility and Management

In the current system, although share-based access is implemented, there is no dedicated mechanism for users to clearly view which files have been shared with them. This limits usability and user awareness. Future enhancements can include a share management dashboard that allows users to:

- View all files shared with them
- Track access permissions
- Receive notifications when a file is shared
- Manage or revoke access rights

Such improvements would significantly enhance the user experience and make the system more practical for collaborative environments.

Scalability and Cloud Integration

The system can be extended to support large-scale deployments by integrating cloud-based infrastructure and distributed computing techniques. This would enable handling of higher workloads and larger datasets efficiently.

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SECURE FILE SHARING SYSTEM USING VISUAL CRYPTOGRAPHY AND BLOCKCHAIN

Prof. Rohan B Kokate[1], Chintu M Sorte[2]

[1]Department Of Computer Applications, JD College of Engineering & Management, Khandala, Borgaon Phata, Kalmeshwar Road, Nagpur, Maharashtra, India – 441501

[2]Department Of Computer Applications, JD College of Engineering & Management, Khandala, Borgaon Phata, Kalmeshwar Road, Nagpur, Maharashtra, India – 441501

1rohan.kokate1@gmail.com, sortechintu310@gmail.com

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ABSTRACT

In the modern digital era, secure file sharing has become a critical concern due to increasing threats to data privacy and integrity. This project, "Secure File Sharing System Using Visual Cryptography and Blockchain," proposes a robust approach that combines the strengths of visual cryptography and blockchain technology to ensure end-to-end file security. The system encrypts a file by dividing it into multiple visual shares using visual cryptography, making each share meaningless on its own. These shares are then distributed and stored securely using blockchain's decentralized ledger, which guarantees immutability, transparency, and resistance to tampering. Users can reconstruct the original file only when all required shares are verified and combined, thereby preventing unauthorized access. Additionally, blockchain smart contracts are employed to manage access control and verify authenticity during file retrieval. This hybrid model enhances data confidentiality, integrity, and trust in file-sharing environments, making it suitable for secure communication in academic, governmental, and corporate domains.

Keywords: Visual Cryptography, Blockchain, Secure File Sharing, Data Privacy, Data Integrity, Decentralized Storage, Smart Contracts.

1. INTRODUCTION

In today's interconnected digital world, the exchange of information and files over the internet has become an integral part of personal, organizational, and governmental operations. However, the rise of cyber threats, data breaches, and unauthorized access has made data security a primary concern. Traditional encryption methods, while effective, often rely on centralized systems that can be vulnerable to hacking, manipulation, or single points of failure. To overcome these limitations, this project introduces a novel Secure File Sharing System that integrates Visual Cryptography and Blockchain Technology to achieve a higher level of security, transparency, and reliability in digital file exchange.

Visual Cryptography (VC) is a cryptographic technique that divides an image or file into multiple shares in such a way that no single share reveals any information about the original data. The original file can only be reconstructed when a sufficient number of shares are combined. This makes it nearly impossible for an attacker to extract meaningful information without access to all the required shares. VC provides simplicity, data confidentiality, and human visual decoding, making it highly effective for secure data transmission.

Blockchain, on the other hand, is a decentralized, distributed ledger technology that ensures data integrity, immutability, and transparency. Each transaction in the blockchain is cryptographically linked to the previous one, creating a tamper-proof chain of records. By leveraging blockchain in this project, file shares can be securely stored and verified without relying on a centralized server. Smart contracts are employed to automate access control and authentication processes, ensuring that only authorized users can request or reconstruct the original file.

The proposed system combines these two technologies to create a secure file sharing environment where files are first divided into visual shares using visual cryptography and then stored in a decentralized manner on the blockchain or through a blockchain-linked storage network such as IPFS (Interplanetary File System). During retrieval, the system verifies the authenticity of each share using blockchain records before reconstruction, ensuring that no unauthorized modification or interception occurs.

In summary, this project aims to design and implement a Secure File Sharing System that utilizes the visual cryptography technique for data encryption and blockchain technology for secure, decentralized verification and storage. By merging these two powerful technologies, the system enhances both data confidentiality and trust, contributing to the advancement of secure communication in the digital age.

2. PROPOSED SYSTEM

2.1 System Architecture

The system is designed as a secure and decentralized file-sharing architecture that combines encryption, visual cryptography, IPFS, and blockchain.

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When a file is uploaded, it is first encrypted (AES) and then split into multiple shares using visual cryptography. These shares are stored separately on a decentralized storage system (IPFS), ensuring that no single location contains the complete file.

The backend acts as an orchestrator, handling validation, hashing (SHA-256), communication with the cryptography service, database transactions, and blockchain interaction.

A relational database stores metadata such as file details and share references, while the blockchain records immutable information like file hash and share identifiers to guarantee integrity and traceability.

This architecture ensures:

- Confidentiality (encryption + splitting)
- Integrity (hashing + blockchain)
- Decentralization (IPFS storage)

Overall, the system eliminates single points of failure and provides a secure, tamper-resistant file-sharing mechanism

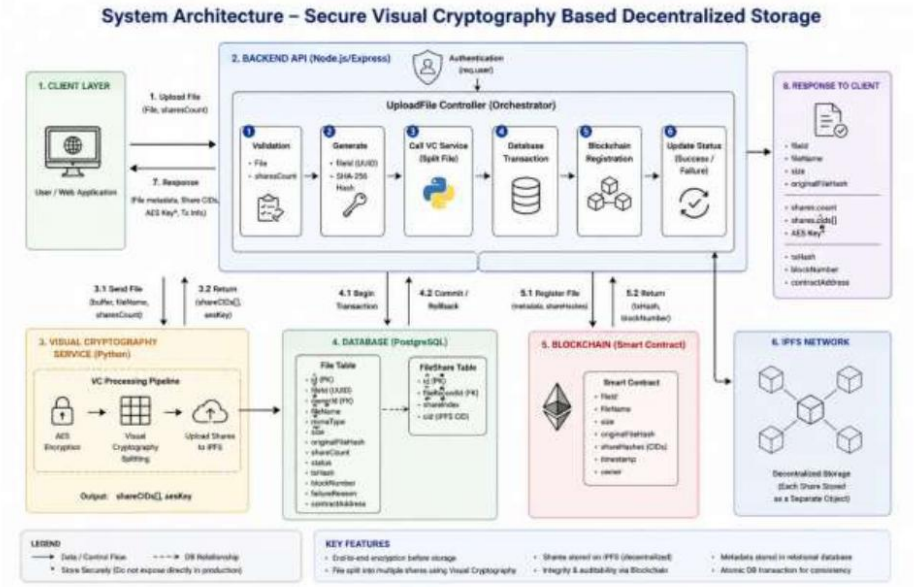

Fig. 1: Proposed System Architecture for Secure Visual Cryptography Based Decentralized Storage

2.2 System Components

The proposed system consists of the following key components:

Client Interface

Provides a user-friendly interface for uploading and retrieving files. It communicates securely with the backend through API requests.

Backend Server (Orchestrator)

Acts as the central control unit responsible for:

- Request validation and authentication
- File hashing (SHA-256)
- Coordination with cryptographic services
- Managing database transactions
- Interacting with the blockchain

Cryptography Service

A dedicated module responsible for:

- Encrypting files using AES
- Splitting encrypted data into multiple shares using visual cryptography
- Preparing shares for decentralized storage

IPFS (Decentralized Storage)

Stores individual file shares as separate entities using content-based addressing (CIDs), ensuring distributed and fault-tolerant storage.

Database

Maintains metadata including file details, ownership, share references, and system status to ensure consistency and efficient retrieval.

Blockchain Network

Stores immutable metadata such as file hash and share identifiers through smart contracts, ensuring integrity, transparency, and auditability.

2.3 System Workflow

The system follows a structured workflow for secure file handling:

File Upload:

The user uploads a file along with required parameters.

Validation & Hashing:

The backend validates input and generates a unique file ID and SHA-256 hash.

Encryption & Splitting:

The file is encrypted using AES and split into multiple shares using visual cryptography.

Decentralized Storage:

Each share is uploaded to IPFS, generating unique CIDs.

Metadata Storage:

File and share details are stored in the database using a transactional approach.

Blockchain Registration:

File metadata and share references are recorded on the blockchain via smart contracts.

Response Generation:

The system returns file details, share references, and transaction information to the user.

3. IMPLEMENTATION

This section describes the practical implementation of the proposed secure file-sharing system, focusing on core modules, algorithms, API structure, and system integration.

3.1 Module Description

The system is divided into three primary modules:

Upload Module:

Handles file upload, validation, hashing, encryption, and splitting into shares.

Share Management Module:

Stores generated shares on IPFS and maintains their references (CIDs) in the database.

Reconstruction Module:

Retrieves required shares from IPFS and reconstructs the original file using decryption and merging logic.

3.2 Algorithms Used

AES Encryption:

The uploaded file is encrypted using Advanced Encryption Standard (AES) to ensure confidentiality before processing.

Visual Cryptography / Shamir's Secret Sharing:

The encrypted file is divided into multiple shares such that only a sufficient number of shares can reconstruct the original data.

SHA-256 Hashing:

A cryptographic hash of the original file is generated to ensure data integrity and is later verified using blockchain records.

3.3 API Design

The system exposes RESTful APIs for handling file operations.

Upload API:

Accepts file input and share count, processes encryption and splitting, and stores metadata.

Reconstruction API:

Fetches required shares and reconstructs the original file.

Metadata API:

Retrieves file details, share references, and blockchain transaction data.

3.4 Integration

The system ensures strong security through multiple layers. Encryption using AES protects file content from unauthorized access. Splitting the file into shares ensures that no single entity can access complete data. SHA-256 hashing combined with blockchain storage guarantees data integrity and prevents tampering. The decentralized nature of IPFS further eliminates single points of failure.

4.4 Comparison with Existing Systems

Metric	Proposed System	Traditional System
Storage	Distributed (IPFS)	Centralized
Security	High (Encryption + Splitting)	Moderate
Integrity	Blockchain-based	Limited
Fault Tolerance	High	Low
Data Privacy	Strong	Moderate

Overall, the results demonstrate that the proposed system provides improved security, integrity, and decentralization compared to conventional file-sharing methods, with a reasonable trade-off in performance and storage overhead.

5. CONCLUSION

This project presents a secure and decentralized file-sharing system that integrates encryption, visual cryptography, IPFS, and blockchain technologies. The system addresses the limitations of traditional centralized storage by ensuring that files are not stored in a single location and are protected through encryption and share-based distribution.

The proposed approach successfully enhances data confidentiality, integrity, and reliability by combining cryptographic techniques with decentralized storage and immutable blockchain records. Overall, the system provides a secure, tamper-resistant, and decentralized solution for modern file-sharing requirements.

6. FUTURE WORK

The proposed system can be further enhanced in several areas to improve security, scalability, and usability. A key improvement is in secure key management, where the AES key should be protected using techniques such as public-key encryption or secure key storage instead of being exposed directly. The system can also be optimized by implementing efficient threshold-based reconstruction, reducing the number of shares required and improving reconstruction speed.

Additionally, access control mechanisms using smart contracts can be introduced to enable fine-grained, permission-based file sharing. Performance can be further improved by optimizing encryption, share generation, and IPFS interactions to handle larger files more efficiently. Finally, enhancements in UI/UX design can make the system more user-friendly and accessible, improving overall user experience

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The system integrates multiple components to achieve secure and decentralized storage:

- The backend server coordinates all operations and communicates with the cryptography service.
- The cryptography module performs encryption and share generation.
- Generated shares are uploaded to IPFS, returning unique CIDs.
- File metadata and share mappings are stored in the database.
- The blockchain records file hash and share identifiers to ensure immutability and traceability.

This integration ensures that the system operates securely, with no single point storing the complete file, thereby enhancing confidentiality and reliability.

4. RESULT & ANALYSIS

This section presents the evaluation of the proposed system in terms of performance, security, and comparison with traditional file-sharing approaches.

4.1 Output Screenshots

The system was tested for file upload, share generation, IPFS storage, and reconstruction. The outputs confirm that files are successfully encrypted, split into shares, and reconstructed without data loss.

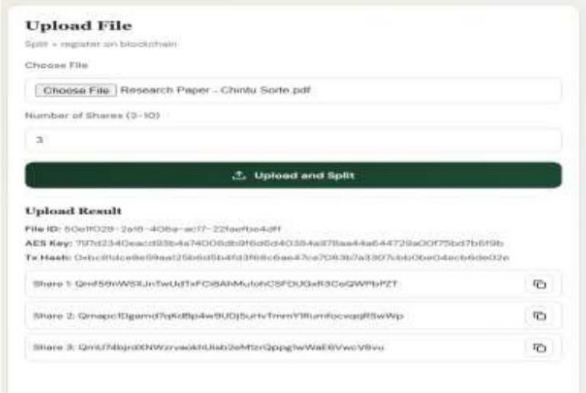

Fig 2. Output Screenshot showing

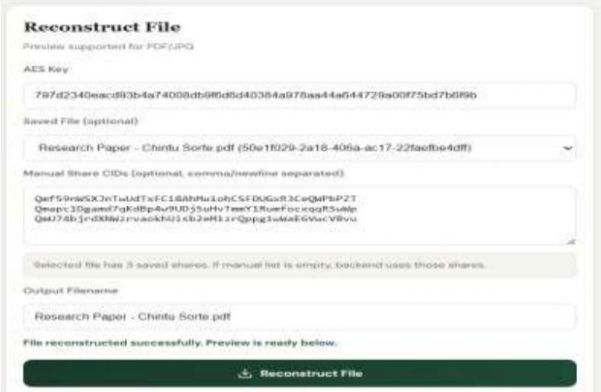

Fig 3. Output Screenshot showing reconstructed generated AES Key and Shares for file using AES Key and share CIDs.

the uploaded image.

4.2 Performance Analysis

The performance of the system was evaluated based on processing time and storage overhead. It was observed that the time required increases with file size and number of shares due to encryption and splitting operations. However, the system maintains acceptable performance for moderate file sizes. Storage overhead is slightly higher compared to traditional systems due to multiple share generation, but this trade-off improves security and reliability.

4.3 Security Observations

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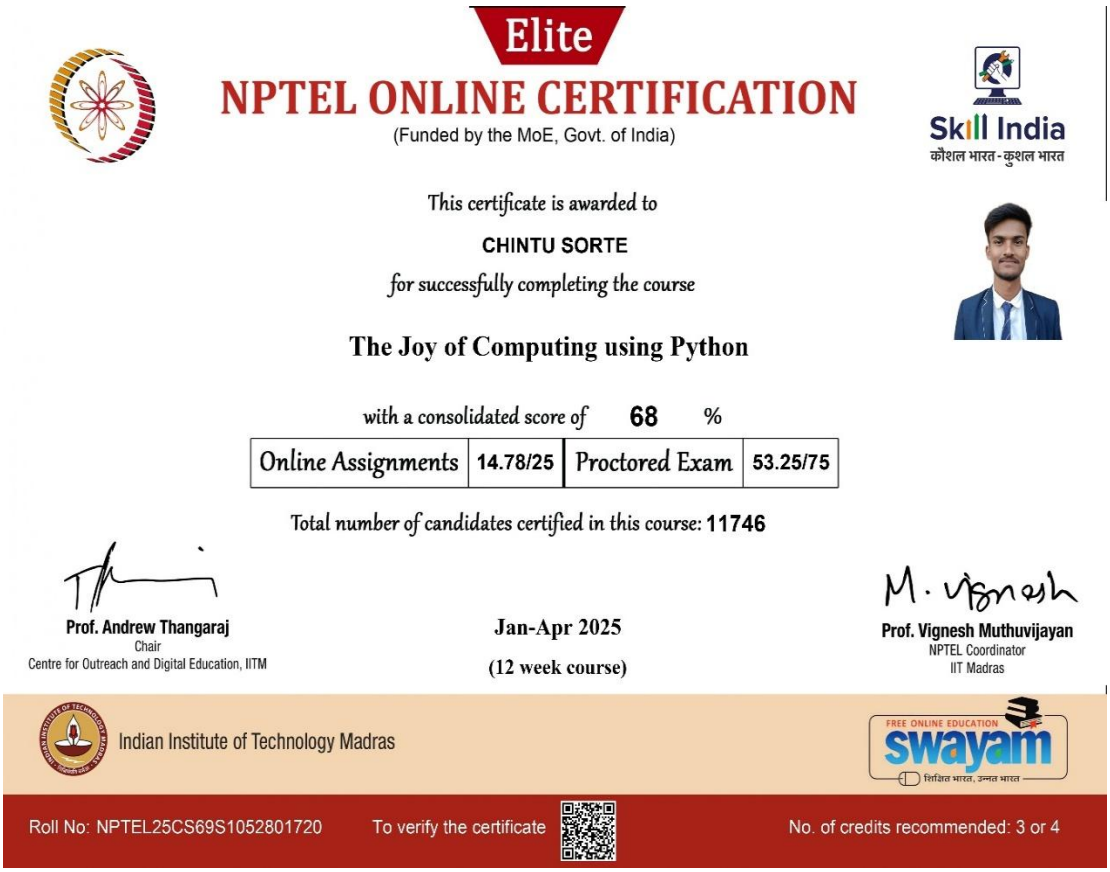

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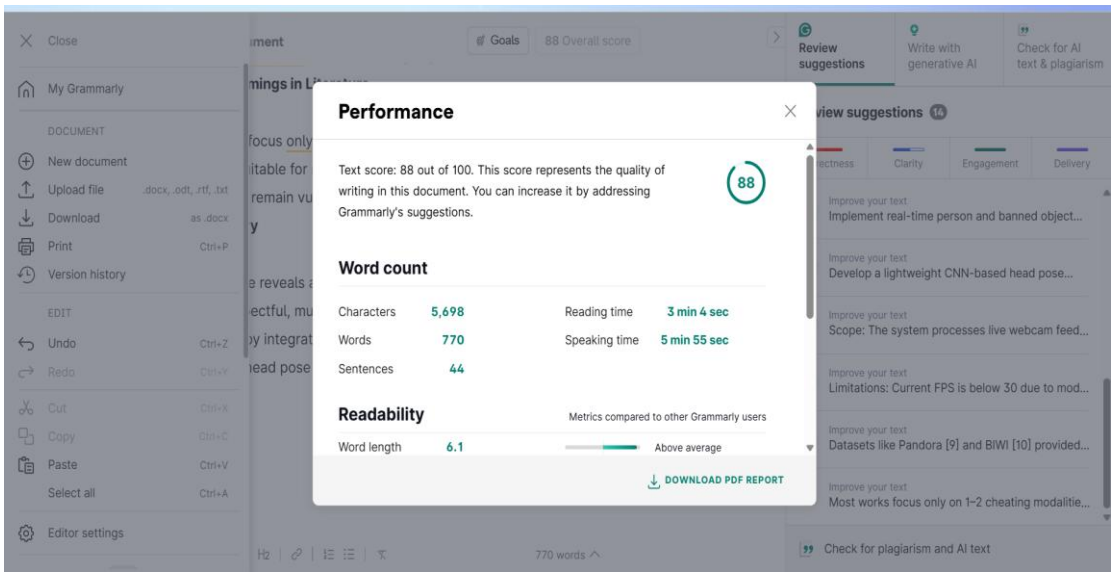

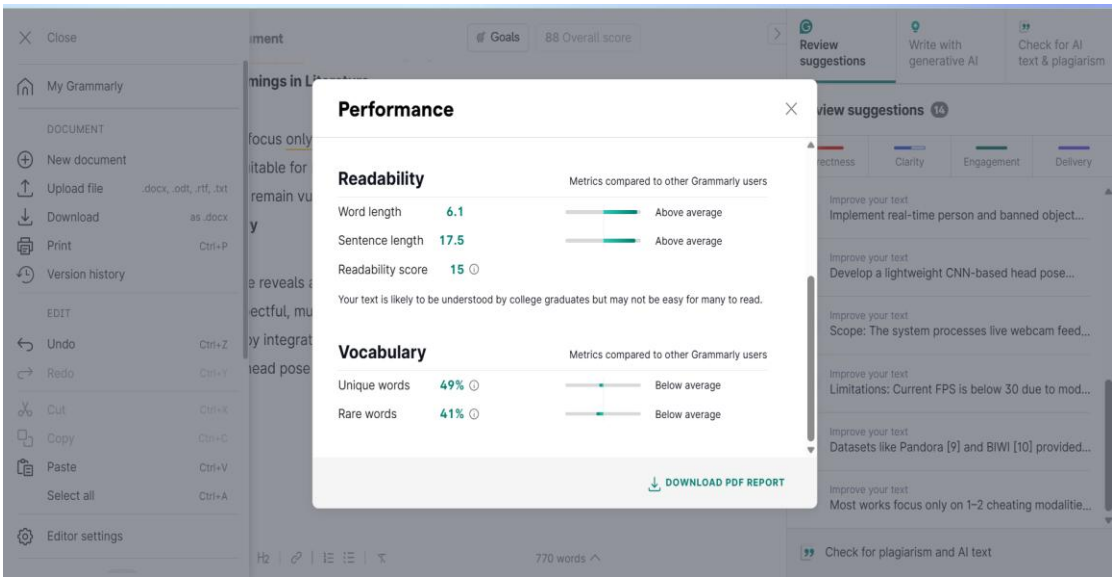

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Mr. Chintu Sorte is currently pursuing a Master of Computer Applications at Dr. Babasaheb Ambedkar Technological University (DBATU), where he is developing strong expertise in full-stack development, application architecture, and secure systems.

He is passionate about exploring modern technologies and actively works on innovative projects involving React, Firebase, Node.js, MongoDB, blockchain, IPFS, PostgreSQL and visual cryptography. His interest lies in building practical, scalable, and impactful digital solutions that integrate security, efficiency, and user-centric design.

Beyond academics and technical work, Chintu is a creative individual with a strong passion for sketching—especially graphitic art—along with interests in web development, reading, volunteering, and continuous learning. His enthusiasm for improving communication skills, exploring new tools, and expanding his knowledge reflects his commitment to personal and professional growth.

He aims to contribute to real-world projects that combine technology, creativity, and meaningful social impact.